

# HUANXUAN LIAO

🏠 [xnhyacinth.github.io](https://xnhyacinth.github.io)

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## EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA) 2023 – Present

PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: [Shizhu He](#) and [Kang Liu](#).

North China Electric Power University (NCEPU)

2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Rank: 1/13.

## RESEARCH INTERESTS

I work toward foundation models for **long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

**Long-Context Data & Model Co-Design**: long-context data curation and synthesis, length- and capability-aware mixtures, scalable pretraining and mid-training, efficient attention, KV-cache optimization, and model architectures to jointly improve long-context capability, scalability, and efficiency.

**Knowledge & Continual Adaptation**: acquiring, transferring, internalizing, and updating knowledge and capabilities across evolving tasks, balancing plasticity, retention, and generalization.

**Agent Learning & Self-Improvement**: persistent memory, long-horizon decision-making, and learning from interaction and feedback to verifiably refine agent memory, policies, and learning procedures.

## SELECTED PUBLICATIONS AND WORKING PAPERS

Full publication list: [personal homepage](#).

### Long-Context Data and Model Co-Design

- [12] [ACM TOIS'26] [Huanxuan Liao](#), Yao Xu, Shizhu He, Jun Zhao, and Kang Liu. “HyCo<sub>2</sub>: Hybrid Context Compression with Global Semantics and Local Evidence”. *Accepted*.
- [11] [arXiv'26] [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, and Kang Liu. “ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning”.
- [10] [AAAI'26] [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, and Kang Liu. “Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning”.
- [9] [arXiv'25] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), et al. “A Comprehensive Survey on Long Context Language Modeling”. *Core contributor to the survey*.  
**Awesome-LCLM**: [Curator & Maintainer](#).

### Knowledge and Continual Adaptation

- [8] [ACL'26] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, and Kang Liu. “Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs”.
- [7] [AAAI'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, and Jun Zhao. “Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks”.
- [6] [COLING'25] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, and Kang Liu. “SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models”.
- [5] [COLING'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, and Jun Zhao. “Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering”.
- [4] [NeurIPS'24] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, and Jun Zhao. “From Instance Training to Instruction Learning: Task Adapters Generation from Instructions”.

## Agent Learning and Self-Improvement

- [3] [EMNLP'26] Yuqiao Tan, Minzheng Wang, Shizhu He, [Huanxuan Liao](#), Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, and Kang Liu. "Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies". **Oral**.
- [2] [ACL'26] Yupu Hao, Zhuoran Jin, [Huanxuan Liao](#), Kang Liu, and Jun Zhao. "Pushing the Limits of LLM Tool Calling via Experiential Knowledge Integration and Activation". *Findings*.
- [1] [ACL'25] Yupu Hao, Pengfei Cao, Zhuoran Jin, [Huanxuan Liao](#), Yubo Chen, Kang Liu, and Jun Zhao. "Evaluating Personalized Tool-Augmented LLMs from the Perspectives of Personalization and Proactivity".

## SELECTED AWARDS

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- National Scholarship (three awards). 2021 / 2022 / 2025
- Beijing Outstanding Graduate. Jun. 2023

## INTERNSHIP AND PROJECT EXPERIENCE

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**Tencent (Project UP), Beijing** Jul. 2026 – Present  
Pretraining intern: long-context data curation and mixture design, scalable sparse attention, and vision-code pretraining data.

**ByteDance, Beijing** Jun. 2025 – Jun. 2026  
Research intern in efficient multimodal modeling: visual budget allocation (ResAdapt), production AIGC detection, and long-horizon agent evaluation (OPERATE).

**Ant Group, Beijing** Feb. 2025 – May 2025  
Research intern in long-context modeling: hybrid context compression (HyCo<sub>2</sub>), the LCLM-Horizon survey, and curation and maintenance of Awesome-LCLM.

## PROFESSIONAL SERVICES

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**Conference Reviewer:** ACL ARR 2024–2026; NeurIPS 2025–2026; ICML 2026; COLM 2026; ICLR 2025–2026; AAAI 2026.

**Open Source:** [Awesome-LCLM](#), **Curator & Maintainer**; [NVIDIA/kvpress](#), contributor; [Distill Yourself](#), agent memory and preference distillation.